

Technical Datasheet RS485 Isolator-Repeat Converter

An **RS485-to-RS485 isolator-Repeater** is a specialized communication interface module designed to **electrically isolate** two independent RS485 networks while still allowing seamless bidirectional data transfer between them. RS485 is widely used in industrial automation, Modbus RTU systems, building management, and long-distance data communication. However, RS485 networks are often exposed to problems such as noise, voltage differences between grounds, and lightning surges.

To overcome these issues, an RS485 isolator provides **galvanic isolation**, ensuring high reliability, safety, and long-term stability.



(RS485-Repeater)

Note – This image is for representation only. Actual product detail may slightly vary.

Specifications -

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw Terminal Block
Dimensions	110 mm L x 50 mm B x 50 mm H
Power	Input Power – 12 – 24 DC Typical – 12V DC @ 80mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing.

Certifications



Additional Features: -

All inputs and communication port isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, □ 30KV contact, □ 30KV air EFT IEC 61000-4-4, 50A (5/50ms) 750V– 5KV isolation.

CRC Error check.

No configuration needed on the IO board

Features:-

- Galvanic Isolation: (typically 750V – 5 kV isolation)
- Bidirectional half-duplex RS485 communication.
- Opto coupler isolation: between input and output signals.
- Isolated DC-DC power module: to supply the RS485 side.
- Protection from ground loops & lightning surges.
- Improves stability for long-distance communication.
- Works with Modbus RTU, DMX, industrial RS485 devices.
- It support wide range of baudrate 9600-2Mbps.
- Transmission Distance up to 1200meters(at 9600 baud)
- 300 meter at high speed like 115200 baud-rate.

Typical Components on the Board:-

- **RS485 Transceiver (e.g., MAX485 / SP485 / ADM485)** on both sides
- **Digital isolator / Optocoupler** for TX/RX signals
- **Isolated DC-DC converter** (e.g., 5V → 5V isolated)
- **TVS diodes / protection circuitry • Terminal blocks** for both RS485 ports.
- **Status LEDs** (Power, TX, RX).

Application:-

1. Industrial Automation:

- PLC communication.
- VFDs ,meter, sensor.
- SCADA system.
- Modbus Network.

2. Long-Distance Communication Lines:

- Between buildings.
- Across large factory.

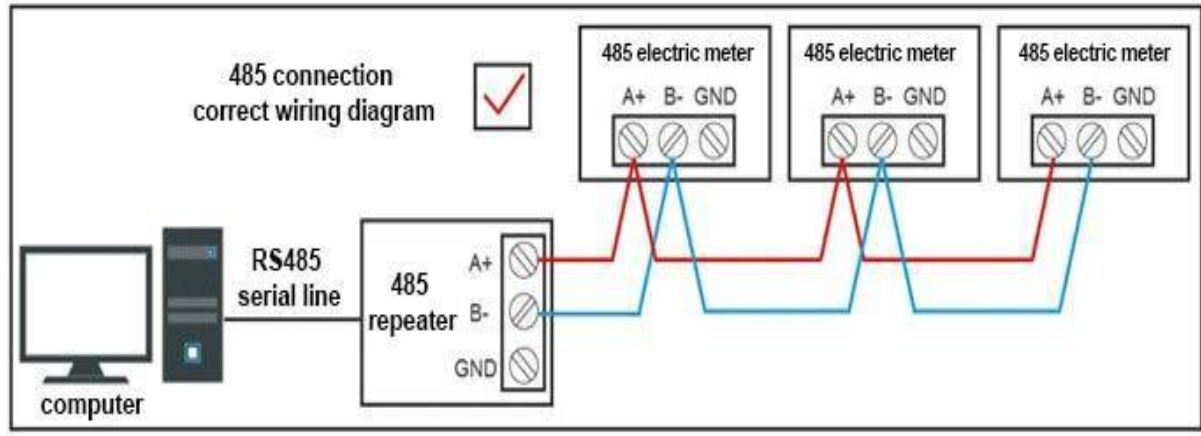
1. Safety-Critical Application:

- Protective expensive equipment.
- Preventing ground Potential difference.

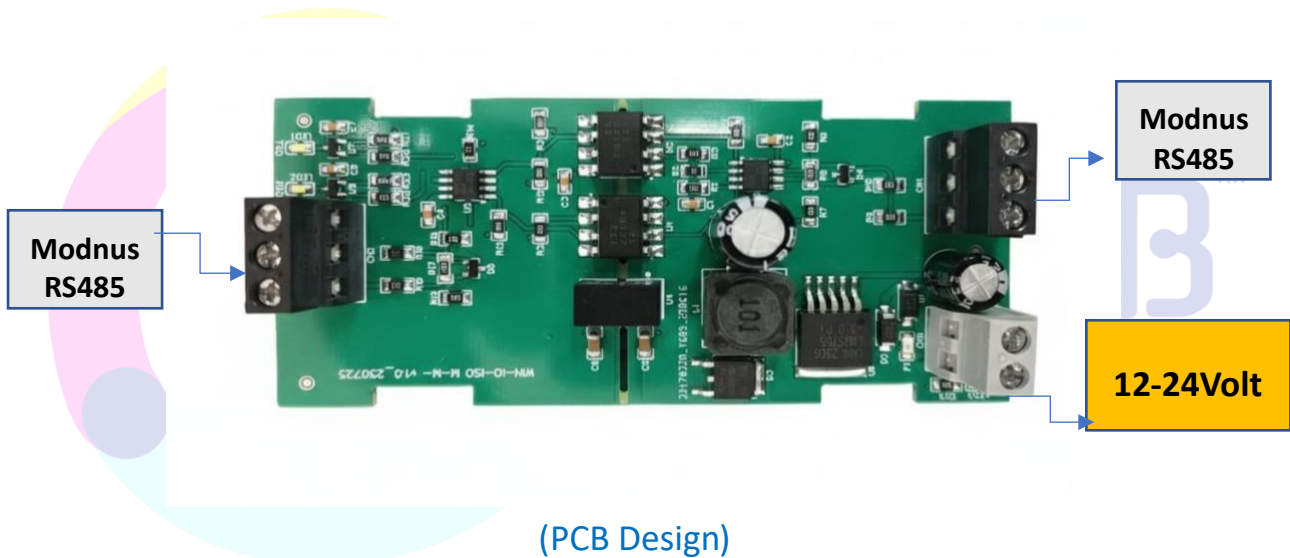
2. Noise Heavy Environments:

- High-Power motors.
- Inductive loads.
- Lightning-prone outdoor installation.

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(Wiring Diagrams)



(PCB Design)

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